Clastify

Investigation Title: An investigation into the effect of the frontal area-to-mass ratio (A/m) of a pendulum bob on the damping rate of its oscillations in air

Subject: Physics SL
Session: May 2026

Student Code: mjs-190

Introduction

Oscillations are a fundamental part of mechanics, appearing in systems ranging from pendulum clocks to the suspension in cars and even the vibrations of atoms in solids. In an idealised case, a pendulum undergoing simple harmonic motion (SHM) would swing indefinitely with constant amplitude. However, in reality, energy is gradually **transferred** to the surroundings through frictional forces, causing the oscillations to diminish over time. This reduction in amplitude is known as **damping**. One of the main causes of damping in everyday pendulums is **air resistance**, which converts mechanical energy into **thermal energy** and sound in the surrounding air.

Understanding damping is important because it helps explain how real oscillatory systems behave and why they eventually come to rest. Applications of damping can be seen in **engineering** (e.g. car suspension systems), **architecture** (e.g. shock absorbers in bridges and skyscrapers), and **technology** (e.g. vibration control in precision instruments), areas where managing oscillations is crucial for performance and safety.

The specific factor being studied in this investigation is the effect of the **frontal (cross-sectional)** area-to-mass ratio (A/m) of a pendulum bob on the rate at which its oscillations are damped by air resistance. Larger frontal areas increase the resistive drag force acting on the pendulum, while larger masses increase the inertia opposing that force.

Clastify

Therefore, the ratio of area to mass provides a meaningful way to compare bobs of different shapes and sizes. This relationship is not only relevant for pendulums but also in real-world contexts: for example, in parachute design (a large area relative to mass produces greater air resistance and slower descent), or in sports equipment such as shuttlecocks and balls, where shape and mass affect how quickly motion slows in air.

On a personal level, I chose this topic because I find oscillations both intuitive and visually engaging. A pendulum is a simple system that allows abstract physics concepts such as damping and energy dissipation to be observed directly. At the same time, the experiment remains feasible using accessible materials, such as everyday objects of different shapes and masses.

Globally, investigating damping has significance in designing safer and more efficient systems, from transport to construction, since controlling oscillations is often essential for stability.

The device used for this experiment is a **simple pendulum**, consisting of a string fixed at one end with a mass (the bob) attached at the other. The pendulum was chosen because it provides a straightforward and controlled way to study oscillatory motion. The setup is versatile enough to allow different bobs to be tested, while the motion is easily measurable using video analysis. This makes the pendulum an appropriate and effective system for investigating how the area-to-mass ratio affects damping, while keeping the experiment within the scope of the IB Physics SL syllabus.

Clastify

Research Question

The aim of this investigation is to determine:

“To what extent does the frontal (cross-sectional) area-to-mass ratio (A/m) of a pendulum bob affect the damping rate of its oscillations in air, as measured using video analysis of peak amplitudes?”

In this experiment, the **independent variable** is the frontal area-to-mass ratio (A/m) of the pendulum bob, expressed in $m^2\ kg^{-1}$.

The **dependent variable** is the damping rate (γ), expressed in s^{-1}.

Clastify

Background Theory

Oscillations such as those of a pendulum are often modelled as **simple harmonic motion (SHM)**, where the restoring force is proportional to the displacement from equilibrium. For small angular displacements (typically less than 10°), the pendulum obeys the approximation:

$$T \approx 2\pi\sqrt{\frac{L}{g}} \qquad [1]$$

where T is the period of oscillation (s), L is the length of the pendulum (m), and g is the acceleration due to gravity. In this idealised case, the pendulum would swing indefinitely with constant amplitude.

In reality, oscillations gradually decrease in amplitude due to **damping** - the process by which mechanical energy is **transferred** from the system to the environment. For a pendulum moving in air, the main damping force is **air resistance (drag)**. This resistive force depends on the **frontal (cross-sectional) area** of the pendulum bob exposed to airflow, as well as its **velocity** and **mass**. Larger frontal areas experience greater drag, while larger masses have higher inertia that resists deceleration. Consequently, the **frontal area-to-mass ratio (A/m)** is the key factor influencing the damping rate.

When damping is small, the amplitude of oscillation decreases approximately **exponentially** with time:

$$x(t) = x_0 e^{-\gamma t} \qquad [2]$$

where $x(t)$ is the instantaneous amplitude (or angular displacement), x_0 is the initial amplitude, t is time, and γ (s^{-1}) is the **damping constant**, which characterises how quickly

Clastify

the amplitude decreases. Larger values of γ indicate faster damping. In typical low-damping systems such as air-damped pendulums, γ values are on the order of **10^{-3} to 10^{-1} s^{-1}**, depending on the bob's size and mass.

If air resistance is approximately proportional to the bob's velocity, a valid assumption for small angles and low speeds, then the damping constant is expected to scale as:

$$\gamma \propto \frac{A}{m} \qquad [3]$$

This relationship provides the theoretical basis for the investigation: bobs with a larger **area-to-mass ratio (A/m)** should exhibit a steeper exponential decay and therefore a larger damping constant γ.

The damping constant can be determined experimentally by recording the pendulum motion and analysing the decay of its **peak amplitudes**. By plotting **ln(x)** (where x is the peak angular displacement in radians) against time t, the data should form a straight line with gradient equal to $-\ \gamma$:

$$ln(x) = ln(x_0) - \gamma t \qquad [4]$$

This analytical method allows precise determination of the damping constant using **video analysis software (Tracker)**, which minimises reaction-time errors and provides consistent frame-by-frame measurements of displacement. Compared to timing full oscillations with a stopwatch, analysing amplitude decay using video produces far more reliable data and enables clear identification of exponential trends.

This forms the theoretical basis for comparing γ across pendulum bobs with different **area-to-mass ratios (A/m)**.

Clastify

Variables

Independent Variable

- Frontal (approximate cross-sectional) area-to-mass ratio (A/m) of the pendulum bob, measured in $m^2\ kg^{-1}$.
- This variable was changed by using ten bobs of varying frontal areas and masses (5 cent coin, 1 cent coin, 20 g weight, 50 g weight, 100 g weight, CD disc, plastic bottle cap, metal washer, plastic jar lid, and small candle base).
- Range of A/m ratio: approximately $1.1 \times 10^{-2}\ m^2\ kg^{-1}$ – $7.3 \times 10^{-1}\ m^2\ kg^{-1}$
    - This range was selected because it allows clear differences in damping to be observed without causing the pendulum to stop too quickly (at very high A/m) or oscillate for excessively long durations (at very low A/m).
    - All values are achievable with common materials and ensure that the pendulum completes at least 15 oscillations before the amplitude decays significantly, making analysis feasible.

Dependent Variable

- **Damping constant** γ, measured in s^{-1}.
- Obtained by recording pendulum motion with a smartphone camera (≥60 fps, ±0.017 s timing resolution) and analysing peak amplitudes using Tracker software.
- The damping constant is determined from the gradient of the straight line on a plot of ln(A) vs time.
- **Uncertainty in γ:** estimated from the standard error of the regression fit, typically $\pm\ 0.002s^{-1}$ depending on data scatter.

Clastify

Control Variables

Control Variable	How It Will Be Controlled	Why It Needs to Be Controlled
Pendulum length, *L*	Measured from pivot to bob's centre of mass with a ruler (±0.5 mm); kept constant at 1 m	Length affects period and speed of oscillation, which influences air resistance.
Type and thickness of string	Same thin, inextensible string used for all trials	Different strings could stretch or create extra drag, affecting results.
Initial angle of release	Set to 8° each time using a protractor (±0.5°); released without push	Larger angles increase speed, which would make drag forces nonlinear.
Release method	Bob held against a fixed marker at 8° and released without push	Prevents adding extra velocity, ensuring consistent starting conditions.
Camera position and setup	Camera on a shelf, fixed distance from pendulum, perpendicular to swing plane	Misalignment or different perspectives would cause parallax error in amplitude measurements.
Background and scale	High-contrast background with a vertical reference line kept constant	Ensures amplitudes can be consistently measured in video analysis.
Environment (air currents)	Room kept closed, AC is switched off	Air movement would add uncontrolled damping, skewing results.
Bob surface finish/material	Smooth, non-porous surfaces (metal/plastic/cardboard coated)	Surface roughness could change drag force independently of area.

Clastify

Equipment

- **Stand with clamp** – to hold the pendulum string securely in place.
- **Thin inextensible string** (1 m length ± 1 mm) – used to suspend the pendulum bob.
- **Smartphone camera** (≥60 fps) – for recording the pendulum motion.
    - Timing uncertainty: ±0.017 s (at 60 fps, 1 frame).
- **Tripod or stable support** – to keep the camera fixed in position and perpendicular to the swing plane.
- **Digital balance** (± 0.05 g) – for measuring the mass of each bob.
- **Ruler** (Resolution: 1 mm; Uncertainty: ± 0.5 mm) – for measuring string length and bob dimensions.
- **Vernier calipers** (±0.05 mm) – for more precise measurements of the bob diameter, used to calculate frontal area.
- **Protractor** (±0.5°) – for setting the initial release angle of the pendulum (8°).
- **10 objects of different shapes/sizes** (e.g., coins, lids, cardboard discs, small balls) – to act as pendulum bobs.
- **High-contrast background board** – to improve visibility of pendulum motion in video.
- **Books/boxes** – to block drafts and minimise external air currents.
- **Video editing software and spreadsheet software** – for video analysis and data processing. I used Tracker software.

Clastify

Method

1. **Assemble the pendulum apparatus.**

   Fix a clamp to a stand and suspend a thin, inextensible string from it. Ensure the pendulum can swing freely without touching nearby objects.

2. **Measure the pendulum length.**

   Using a ruler (±0.5 mm), measure the distance from the pivot point to the centre of mass of the bob. Keep this length constant at 1.00 m throughout all trials to control for its effect on oscillation period and air resistance.

3. **Prepare the background and camera setup.**

   Place a high-contrast vertical background with a reference line behind the pendulum to make amplitude measurements clearer during video analysis.

   Position a smartphone camera approximately 1.0 m away from the pendulum, supported on a stable shelf, and aligned perpendicularly to the plane of oscillation at the bob's equilibrium height. This setup minimises parallax error and ensures consistent framing for all recordings.

4. **Attach and measure each pendulum bob.**

   Secure the first bob to the end of the string. Measure its **mass** using a digital balance (±0.005 g) and its **diameter or radius** using vernier calipers (±0.05 mm). Calculate its frontal **area-to-mass ratio (A/m)** and record this value.

5. **Set and control the initial release angle.**

   Using a protractor (±0.5°), displace the bob to an initial angle of approximately 8° from the vertical. Hold the bob against the protractor and release it gently without applying any push to ensure identical starting conditions for all trials.

6. **Record the oscillations.**

   Start recording the pendulum motion with the smartphone camera at a frame rate of 60

Clastify

frames per second. Record for at least **90 seconds**, ensuring that at least the **first 15 oscillations** are clearly captured. Only these initial oscillations were used for analysis, as they provided sufficient data before the amplitude decayed significantly. This recording duration was kept constant for all bobs to maintain fair testing conditions.

7. **Control environmental conditions.**

   Conduct the experiment in the school's closed prep room, with doors shut and the air conditioning turned off, to eliminate unwanted air currents that could affect damping.

8. **Track the motion using Tracker software.**

   Import each recorded video into **Tracker** and track the pendulum bob frame by frame. Determine the **angular displacement** at each successive peak, converting the angles from degrees to **radians** for analysis consistency.

9. **Extract and process data.**

   For each trial, record at least 15 successive peak amplitudes. Note the corresponding time for each peak and calculate the natural logarithm of each amplitude $ln(A)$.

10. **Determine the damping constant γ.**

    Plot $ln(A)$ against time for each trial and draw the best-fit straight line. The gradient of the line gives -γ, allowing the damping constant to be determined as γ = -gradient.

11. **Repeat and average results.**

    Repeat the procedure **three times for each of the ten bobs** to reduce random errors. Calculate the average damping constant (γ) and its standard deviation for each bob.

12. **Analyse the relationship.**

    Plot the mean damping constants (γ) against their corresponding area-to-mass ratios (A/m). Draw the best-fit line to determine the overall relationship between damping rate and area-to-mass ratio.

Clastify

Risk Assessment

Risk / Concern	Type (Safety / Ethical / Environmental)	Mitigation Strategy
Pendulum bob detaching and striking someone	Safety	Ensure the string and clamp are tightly secured; check knots before each trial.
Pendulum striking objects or people	Safety	Keep the swing path clear; use light objects as bobs; limit release angle to ~8°.
Retort stand tipping over	Safety	Place on a stable surface; add weights to base if necessary.
Trip hazard from equipment setup	Safety	Arrange string, tripod, and wires neatly; keep experiment area organised.
Misuse of heavy or sharp objects as bobs	Safety	Use small, lightweight, and smooth objects only.
Waste of materials (cardboard, plastic)	Environmental	Reuse household items and school lab equipment (coins, weights); can be used after the experiment.
Ethical concerns	Ethical	No living beings involved; no ethical issues identified.

Clastify

Results

Qualitative Observations

- Bobs with larger frontal area-to-mass ratios (A/m) slowed down significantly faster, with amplitudes halving within about 10 oscillations.
- Bobs with smaller area-to-mass ratios (A/m) values maintained visible oscillations for longer, often over 20 oscillations.
- The amplitude decay appeared smooth and gradual, consistent with exponential damping.
- No irregular jerks or disturbances were visible, suggesting that external drafts were successfully minimised.

Clastify

Processed Data

Bob	Radius / m $(\times 10^{-2})$	Area (A) / m² $(\times 10^{-4})$	Mass (m) / kg $(\times 10^{-3})$	A/m / m²/kg $(\pm 0.001\, m^2/kg)$
5 cent coin	1.06	3.53	3.95	0.0894
1 cent coin	0.81	2.06	2.29	0.0900
20g weight	1.25	4.91	20.1	0.0244
50g weight	1.75	9.62	50.4	0.0191
100g weight	1.90	11.3	99.9	0.0113
CD disc	5.98	112.0	15.3	0.732
Plastic bottle cap	1.47	6.79	1.62	0.419
Metal washer (large)	1.63	8.35	15.5	0.0540
Plastic jar lid	3.79	45.1	6.42	0.703
Small candle base	2.46	19.0	17.9	0.106

Table 1. Physical properties and frontal area-to-mass ratios of the ten pendulum bobs used in the investigation.

Clastify

Raw Data

I have recorded each pendulum bob with a phone camera at 60 frames per second and extracted amplitude values using an analysis tool.

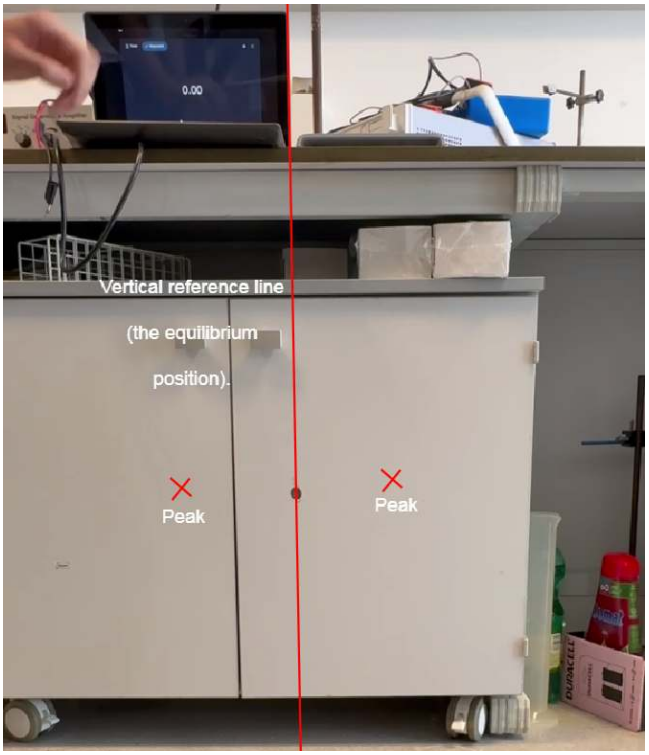


Figure 1. Experimental setup showing the pendulum apparatus with a vertical reference line marking the equilibrium position.

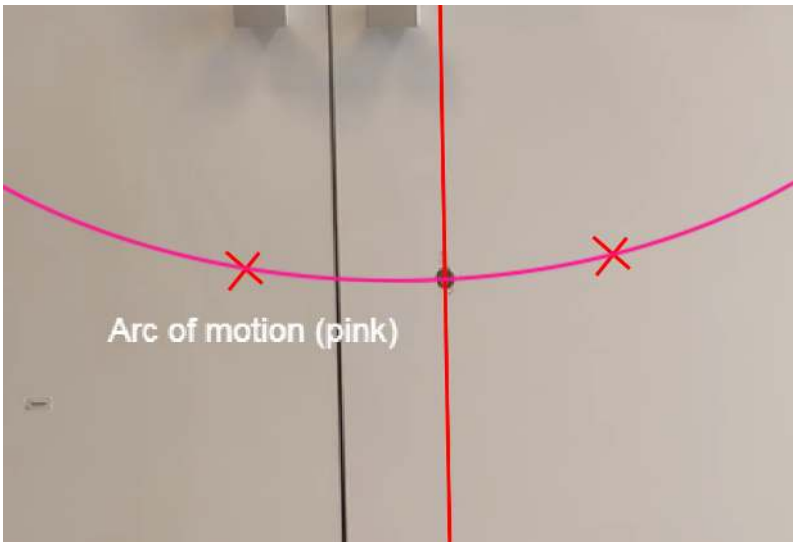


Figure 2. Overlay showing the arc of motion (pink) traced by the pendulum bob and the peak displacement markers (×).

Clastify

Peak Number	Time (t) / s $\pm$ 0.017 s	Amplitude (θ) / ° $\pm$ 0.5°
1	0.50	8.0
2	2.52	7.4
3	4.54	7.4
4	6.55	7.1
5	8.56	7.4
6	10.57	7.2
7	12.58	7.0
8	14.59	6.9
9	16.60	6.7
10	18.61	6.6
11	20.62	6.4
12	22.63	6.3
13	24.64	6.1
14	26.65	5.9
15	28.66	5.8

Table 2. Raw angular displacement data for the 1-cent coin pendulum (L = 1.000 m, initial angle θ_0 = 8.0°).

Time values correspond to successive angular-displacement peaks extracted from 60 fps video footage (timing uncertainty ± 0.017 s; angular uncertainty ± 0.5°).

Clastify

Notes

- Time values correspond to successive peaks on the same side of the swing, separated by approximately one full period ($\sim 2.0\ s$), consistent with a 1 m pendulum ($T \approx 2.0\ s$).
- The amplitude decreased smoothly over 15 oscillations, consistent with exponential decay.
- Data for other bobs (5 cent coin, 20 g weight, 50 g weight, 100 g weight, CD disc, plastic bottle cap, metal washer, plastic jar lid, small candle base) was collected in the same way and processed into damping constants in **Table 5**.

Data Processing

To determine the damping constant γ for each pendulum bob, the exponential decay of amplitude over time was linearised. For a lightly damped oscillation, the amplitude decreases exponentially with time according to:

$$A(t) = A_0 e^{-\gamma t} \qquad [5]$$

Taking the natural logarithm of both sides:

$$ln(A) = ln(A_0) - \gamma t \qquad [6]$$

This means that plotting $ln(\theta)$ against time should produce a straight line with gradient $-\ \gamma$.

Clastify

Example: 1 cent coin (Trial 1)

The raw peak amplitudes from Table 2 were first converted from degrees to radians, then the natural logarithm of each value was calculated using equation [6], as shown in Table 4.

Peak Number	**Time** (t) / s	**Amplitude** θ / °	θ / rad	ln (θ)
1	0.50	8.0	0.140	-1.97
5	8.56	7.4	0.129	-2.05
10	18.61	6.6	0.115	-2.16
15	28.66	5.8	0.101	-2.29

Table 4. Sample processing of raw data for 1 cent coin (Trial 1)

These values were plotted as $ln(\theta)$ vs time, and a best-fit line was fitted to all 15 data points, producing Figure 3.

Clastify

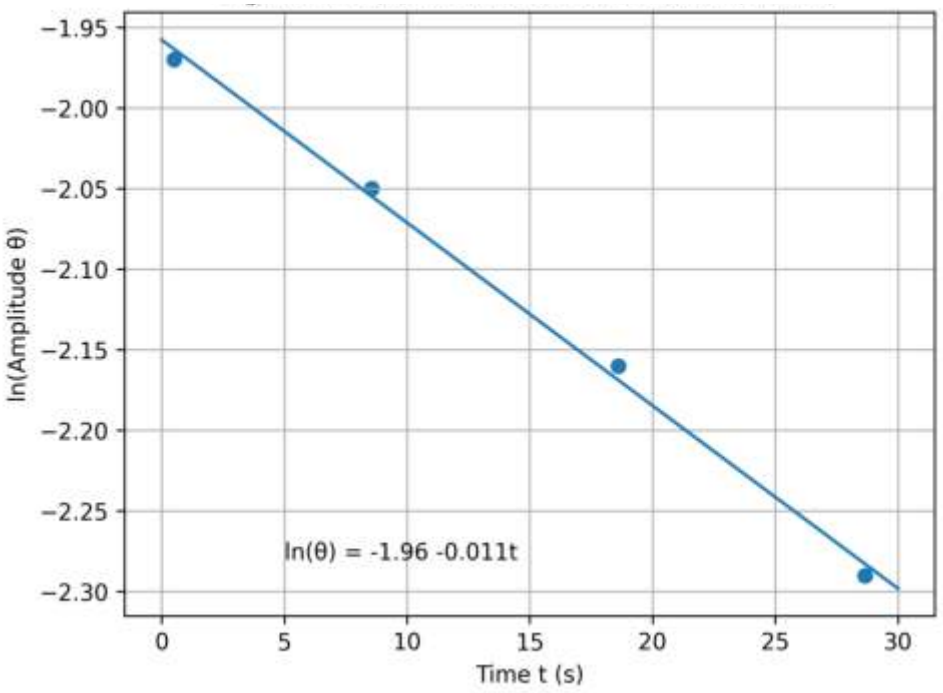


Figure 3. Plot of $ln(\theta)$ against time for the 1 cent coin (Trial 1). The gradient of the best-fit line represents $-\ \gamma$.

From equation [6], the gradient of this line equals $-\ \gamma$, giving:

$$m\ =\ -\ 0.011\ s^{-1} \qquad [7]$$

Therefore:

$$\gamma\ =\ -\ m\ =\ 0.011 s^{-1}$$

Calculating uncertainty across trials (SD)

This procedure was repeated for all five trials of the 1 cent coin. The damping constants obtained were $\gamma\ =\ 0.012, 0.011, 0.013, 0.011, 0.013\ s^{-1}$, giving a mean of $\bar{\gamma} = 0.012\ s^{-1}$. The standard deviation was calculated as:

Clastify

$$SD = \sqrt{\frac{(0.012-0.012)^2 + (0.011-0.012)^2 + (0.013-0.012)^2 + (0.011-0.012)^2 + (0.013-0.012)^2}{5-1}} = 0.0010\ s^{-1}$$

This small standard deviation of $0.0010\ s^{-1}$ indicates high repeatability across all five trials, suggesting that random errors were well controlled.

Full results across all bobs

This linearisation procedure was applied identically to all ten pendulum bobs across five trials each. For every trial, the $ln(\theta)$ vs time graph was plotted, the gradient extracted, and γ calculated using equation [7]. The mean damping constant and standard deviation for each bob are summarised in Table 5.

Clastify

Object	A/m / m²/kg (± 0.001 m²/kg)	γ Trial 1 / s⁻¹	γ Trial 2 / s⁻¹	γ Trial 3 / s⁻¹	γ Trial 4 / s⁻¹	γ Trial 5 / s⁻¹	Average γ / s⁻¹	Standard Deviation γ / s⁻¹
100 g weight	0.0113	0.0020	0.0020	0.0030	0.0020	0.0030	0.0024	0.001
50 g weight	0.0191	0.0030	0.0030	0.0040	0.0030	0.0040	0.0034	0.001
20 g weight	0.0244	0.0030	0.0040	0.0050	0.0040	0.0050	0.0042	0.001
Metal washer	0.0540	0.0070	0.0070	0.0080	0.0070	0.0080	0.0074	0.001
1 cent coin	0.0900	0.011	0.012	0.013	0.011	0.013	0.012	0.001
5 cent coin	0.0894	0.011	0.012	0.013	0.011	0.012	0.012	0.001
Candle base	0.106	0.013	0.014	0.015	0.013	0.014	0.014	0.001
Bottle cap	0.419	0.050	0.051	0.052	0.051	0.052	0.052	0.001
Plastic jar lid	0.703	0.084	0.085	0.087	0.085	0.087	0.087	0.001
CD disc	0.732	0.088	0.089	0.091	0.089	0.090	0.089	0.001

Table 5. Mean damping constants and standard deviations obtained from $ln(\theta)$ vs time plots for all ten pendulum bobs across five trials each. A/m uncertainty of $\pm\ 0.001\ m^2/kg$ is propagated from caliper and balance measurements.

Clastify

Graphical Analysis

Figure 4 shows the relationship between the damping constant γ and the area-to-mass ratio A/m for all ten pendulum bobs tested. Each data point represents the mean damping constant from five trials, with error bars indicating the propagated uncertainties in both γ and A/m, taken from Table 5. The error bars are present but are too small to be visible on the scale of the graph.

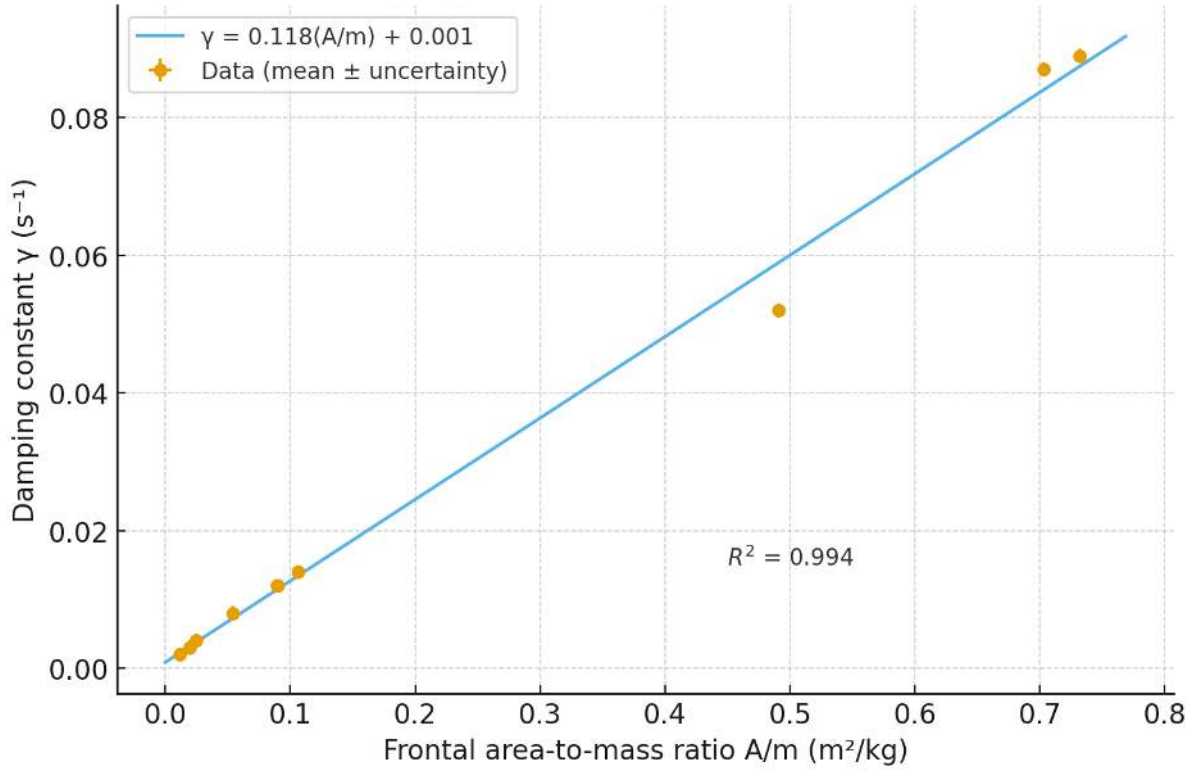


Figure 4. *Damping constant (γ) vs area-to-mass ratio (A/m) for all ten pendulum bobs. Error bars represent propagated uncertainties in both variables. The best-fit line is*

$$\gamma = 0.118\,(A/m) + 0.001,\ with\ R^2 \approx 0.994$$

Clastify

The graph demonstrates a strong **positive linear correlation** between γ and A/m. Therefore, as the frontal area-to-mass ratio increases, the damping constant also increases, meaning that bobs with relatively larger areas (or lighter masses) lose amplitude more quickly. This is consistent with the theoretical prediction that air resistance is proportional to frontal area and inversely proportional to mass, leading to $\gamma \propto A/m$.

The best-fit line has the equation:

$$\gamma = 0.118\,(A/m) + 0.001 \qquad [8]$$

with a coefficient of determination $R^2 \approx 0.994$. The very high R^2 value indicates that the linear model explains nearly all of the variation in the data. The small positive intercept ($0.001\ s^{-1}$) may reflect background sources of damping not directly linked to air resistance, such as internal friction at the clamp or minor energy losses in the string.

Overall, the analysis confirms the hypothesis: **the damping constant is directly proportional to the frontal area-to-mass ratio of the pendulum bob.**

Clastify

Conclusion

This investigation set out to answer the research question:

"To what extent does the frontal area-to-mass ratio (A/m) of a pendulum bob affect the damping rate γ of its oscillations in air?"

The results show a clear and strong linear relationship between γ and A/m. Bobs with the largest A/m values, such as the CD disc $(0.732\ m^2/kg)$, had the highest damping constants of $0.089\ s^{-1}$. By contrast, the heavier weights with much smaller A/m values $(0.0113\ -\ 0.0244\ m^2/kg)$ had damping constants as low as $0.002\ -\ 0.004\ s^{-1}$.

This directly supports the theoretical model that air resistance increases with frontal area and decreases with mass, giving rise to the proportionality $\gamma \propto A/m$. The best-fit line $\gamma = 0.118(A/m) + 0.001$ with $R^2 \approx 0.994$ provides quantitative evidence for this relationship, with the small non-zero intercept likely arising from additional background sources of damping such as internal friction in the clamp and string.

The conclusion is therefore that the **frontal area-to-mass ratio is a key factor in determining the damping rate of a pendulum bob in air**, and the experimental data provides strong evidence that the relationship is linear under the conditions of small oscillations.

Clastify

Evaluation

Strengths

- **Controlled setup:** The pendulum length, release angle, and string type were kept constant throughout, ensuring only the area-to-mass ratio varied.
- **Video analysis:** Using Tracker eliminated human reaction-time errors and allowed precise measurement of angles frame by frame.
- **Multiple trials:** Five runs per bob reduced the effect of random errors, and the use of averages with standard deviations improved reliability.
- **Wide range of A/m values:** The dataset covered nearly an order of magnitude ($0.011–0.732\ m^2/kg$), providing a wide range of damping behaviours and making the linear trend unambiguous.
- **Error analysis:** Propagated uncertainties were explicitly calculated and included in the final graph, showing a high level of rigour.

Limitations

- **Coin rotation:** Flat coin-shaped bobs occasionally rotated during oscillation, slightly altering the effective frontal area and introducing inconsistency.
- **Parallax error:** Although the camera was positioned carefully, small misalignments could still have affected angular readings in Tracker.
- **Environmental factors:** Air currents in the room, though minimised, were not fully eliminated and could have influenced damping.
- **Limited number of data points:** With only ten bobs tested, the dataset was sufficient but not extensive; more objects with a wider range of A/m ratio could strengthen the conclusion.

Clastify

- **Simplified model:** The analysis assumed linear (velocity-proportional) drag, which is only an approximation at low speeds and small angles.

Improvements

- Use **spherical bobs** (e.g., rubber balls, marbles) to avoid rotation issues with flat coins and ensure consistent frontal area.
- Test a **larger sample size** of bobs (15-20 objects) to provide more data points and improve statistical confidence.
- Record at a **higher frame rate** (e.g., 120 fps) to reduce timing uncertainty further.
- Conduct the experiment in a **sealed environment** (e.g., transparent box) to eliminate air drafts.
- Extend the investigation by testing **different pendulum lengths** to confirm whether $\gamma \propto A/m$ holds under varying conditions.

Clastify

Final Reflection

This investigation has shown how a simple setup like a pendulum can reveal the impact of air resistance on oscillatory motion. The clear proportional relationship between the damping constant and the area-to-mass ratio links directly to real-world applications, such as the design of parachutes, vehicles, and sports equipment, where shape and mass strongly affect how objects slow down in air.

On a personal level, carrying out this experiment strengthened my practical and analytical skills. I learned how to control variables carefully, use video analysis software effectively, and propagate uncertainties to produce reliable results. Perhaps the most important lesson was seeing how an idealised concept like simple harmonic motion changes once damping is included, a reminder that real systems rarely behave perfectly, and that experimental design must take this into account.

Overall, this IA not only confirmed the theoretical prediction that $\gamma \propto A/m$, but also deepened my understanding of how physics is applied in both everyday contexts and advanced engineering. It was rewarding to see how a question I was curious about could be answered with a clear experimental approach, and I would like to explore further how these methods could be extended to more complex damping scenarios.

Cite the textbook

Clastify